Aerodynamic Design - Final Report

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Introduction

The lift produced by a wing is limited by its design; however, the lighter that wing is, the greater the plane's useful load will be. A balance must be achieved to maintain sufficient lift under safe conditions, while having the lightest wing possible. For this reason, an analysis was performed to minimize the structural mass of an aluminum wing using the packages `SNOW.jl`, `VortexLattice.jl`, and `GXBeam.jl`. The wing is modeled using parameters similar to those of the Cessna Skyhawk, such as the same freestream velocity and wingspan. It is composed of the same material and must support the same maximum load. The design variables are the chord length, twist, and height distributions, which are used to minimize the mass while meeting the constraints. Some of these constraints are: a maximum allowable bending moment stress, a minimum lift-to-induced-drag ratio, a maximum allowable tip deformation, and a minimum lift force produced (more info can be found in *Optimization Constraints*). The solution is then compared to the Skyhawk's wing to verify that realistic results have been produced.

Methods

The purpose of this analysis is to minimize the mass of the wing while meeting certain constraints (listed below in *Optimization Constraints*), using `VortexLattice.jl`, `SNOW.jl`, and `GXBeam.jl`. As `SNOW.jl` is a single-objective optimizer, only the structural mass is minimized. If a multi-objective optimizer were used, the Lift-to-Induced Drag ratio would also be optimized; however, in this optimization it is used as a constraint (see *Optimization Constraints*). The wing is modeled using parameters similar to those of the Cessna Skyhawk, such as the same freestream velocity and wingspan. It is composed of the same material (Al 6061-T6) and must support the same maximum load. The design variables are the chord length, twist, and height distributions, which are used to minimize the mass while meeting the constraints.

Wing Geometry and Parameters

The wing is modeled as a cantilever beam with a hollow, elliptical cross-section, as shown in Figure 1. Figure 2 shows what some typical airfoil geometries may look like, and they generally have an elliptical shape. For this reason, the airfoil can be approximated as a hollow ellipse, which will provide a similar geometry for `GXBeam` while still being simple to model.

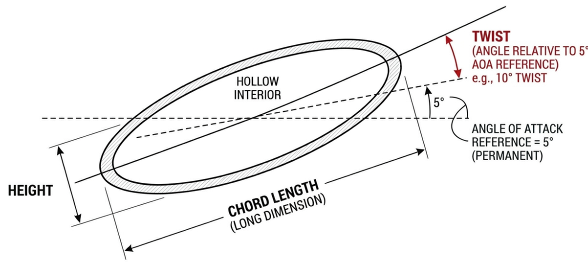

Figure 1: Cross-sectional view of the wing's airfoil used in this model. The height and chord lengths are measured from the outside edges of the airfoil.

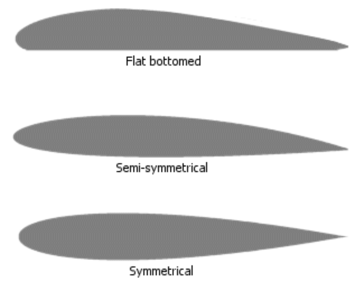

Figure 2: Examples of the cross-sectional geometry of some typical airfoils.[2]

`VortexLattice` does not use this geometry, and it only calculates the lift and drag forces; the total mass and deformation of the wing are found using `GXBeam`, accounting for this cross-sectional geometry.

For simplicity, the wing is modeled without any wing struts (unlike the Cessna Skyhawk), and there are no structural beams inside the wing. The Cessna Skyhawk features structural aluminum beams covered in a thin aluminum sheet (for the wing's skin), as opposed to relying on the thickness of the skin for structural support. The parameters are as follows:

Parameter	**Value**
Wing Span	11m
Freestream Velocity (V_∞)	62.8m/s
Angle of Attack (α)	5°
Spanwise / Chordwise Panels	50 / 4
Air Density (ρ)	1.225 kg/m^3 (Sea Level)
Wall Thickness (Root to Tip)	1.27 cm (0.50 in) – 0.635 cm (0.25 in)

50 spanwise values were used because this quantity yielded results with high accuracy and relatively little noise compared to using fewer panels. No more than 50 panels were used due to the computational cost. The thickness of the wing's wall decreases linearly as it approaches the tip, as shown in Figure 3. These values for the wall thickness were chosen after testing several potential thicknesses. These are the values that allowed the optimization to yield the smallest mass while still meeting the maximum bending moment stress and maximum deformation constraints, as well staying within the height distribution bounds (See *Optimization Constraints* and *Design Variables*).

[2]Source: https://defiantwings.blogspot.com/2017/05/airfoil-airfoil-airfoil.html

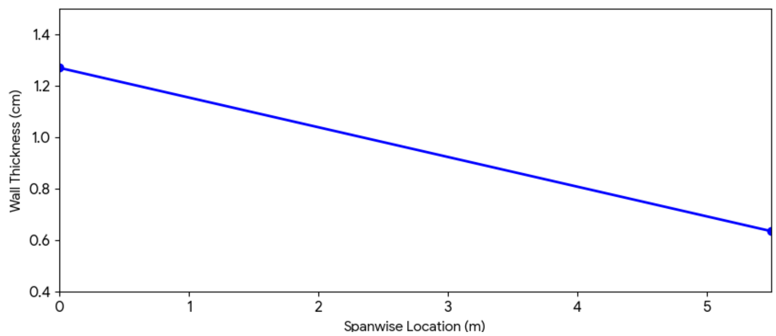

Figure 3: Cross-sectional view of the wing's airfoil. The height and chord lengths are measured from the outside edges of the airfoil.

Design Variables

The optimization utilizes three primary groups of design variables: chord length, twist, and cross-sectional height for each of the 50 spanwise panels used, resulting in a total of 150 design variables.

Variable Group	Lower Bound	Upper Bound	Initial Value
Chord Length (c)	0.2 m	2.5 m	1.0 m
Twist (θ)	$-5°$	$5°$	$0°$
Height (h)	0.01905 m (0.75 in)	0.1270 m (5 in)	0.0254 m (1 in)

The Twist constraints are strictly limited to $\pm 5°$ due to `VortexLattice's` underlying assumption of inviscid airflow. Since the freestream angle of attack is $\alpha = 5°$, any local twist exceeding $5°$ (resulting in an effective $\alpha > 10°$, see Figure 1) would generate inaccuracies in the lift and drag data due to the differences between viscid and inviscid airflow. At these higher angles, flow separation and stall occur, which cannot be accurately modeled using `VortexLattice`.

Optimization Constraints

The optimization is subject to the following conditions:

1. A **'monotonicity'** constraint for the chord, twist, and height distributions, forcing these values to decrease from root to tip.

2. **Lift Requirements:** The full wing must produce a lift force greater than or equal to 11281 N (or support a total mass of 1150 kg), which is the maximum allowable load of a Cessna Skyhawk.

$$Lift \geq 11281\, N \tag{1}$$

3. **Stress Limit:** Must not exceed a maximum bending moment stress of 120 MPa. Al 6061-T6 has a yield strength of 276 MPa at 24° C,[3] which allows a safety factor of 2.3.
$$\sigma \leq 120\, MPa \tag{2}$$
4. **Efficiency:** The Lift-to-Induced-Drag ratio must be greater than 60 to ensure efficiency.
$$\frac{L}{D_i} \geq 60 \tag{3}$$
5. **Deformation:** The maximum deformation in the vertical (z) direction must be no greater than 1m.
$$\delta_z \leq 1\, m \tag{4}$$

1.1 (Monotonicity Constraint) The `Diff()` function finds the difference between two variables (for example, the chord length of panels 1 and 2), and by constraining these differences to always be less than zero, the 'monotonicity' condition forces the design variables to decrease over the length of the wingspan, allowing the optimizer to provide realistic geometry that can be manufactured. Without this, the optimizer provides jagged wing edges, sharp changes in twist (in the positive or negative direction), and the results are difficult to manufacture. With this constraint, there might be drastic changes between panels; however, the distribution curve is generally smoother since it can only decrease and it must stay within the bounds. Additionally, by forcing the twist to decrease over the length of the wing, this monotonicity constraint ensures that the wing will stall at the roots first, rather than the tips. This maintains lateral control and increases stability.

4.1 (Efficiency) `VortexLattice` is used to calculate the induced drag on the wing, and a ratio of Lift-to-Induced-Drag is constrained to be over 60. While `VortexLattice` only calculates induced drag, it is not the only source of drag acting on the wing. Other drag forces such as skin friction, form, and wave drag must be calculated using other methods and are neglected in this study. This ratio is constrained to be 60 to ensure efficiency when the other forms of drag are included. The Cessna Skyhawk has a lift-to-total-drag ratio of 11; however, this discounts all forms of drag other than induced drag. 60 was chosen as the lower bound due to the model's exclusion of the other forms of drag.

5.1 (Deformation) The Cessna Skyhawk has wing struts on both sides of the body,[4] and structural beams inside the wing, which help resist vertical deformation. This greatly helps reduce the mass required to stay within the constraints (specifically, the maximum allowable bending moment stress and the deformation). These internal beams allow specific cross-sections to be chosen for the beams that better resist deflection, such as an I-beam, for example. The wing struts provide a counter-moment opposing the lift force. For simplicity, the optimization's model has the hollow elliptical geometry described above, meaning the minimized mass will be significantly greater than if internal beams were used to support a thinner wall. This is because the current model places mass where it may not be needed.

[3] https://www.matweb.com/search/datasheet.aspx?MatGUID=b8d536e0b9b54bd7b69e4124d8f1d20a
[4] https://cessna.txtav.com/en/piston/cessna-skyhawk

The total structural mass of the wing is calculated by finding the masses of each individual element while creating the mass matrix for `GXBeam`, summing them, and doubling the result. `VortexLattice` and `GXBeam` only analyze a half wing and the results are mirrored when required, requiring the computed mass of the half-wing to be multiplied by 2.

The industry standard for maximum deflection is 5-15% of the semi-span, and this model is at 18%. 1 meter was chosen as the maximum deflection due to the model's lack of wing struts (which help inhibit the deflection). All deformations and twists at the roots are constrained to be zero.

Results and Discussion

The minimized mass is 232.86 kg. The chord, twist, height, and deformation distributions are shown below, as well as their corresponding maximum and minimum values:

Design Variable	**Min Value**	**Max Value**
Chord Length	0.200 m	0.566 m
Twist	$-5°$	$+5°$
Height	2.06 cm	11.75 cm
Deformation	0 m	1 m

The results approached the lower limit for the chord distribution, and hit the upper and lower limits for the twist distributions, the lower limit for the height distribution, and the upper limit for the deformation distribution. This verifies that the constraints used correctly limited the optimization and were needed to provide realistic results. For example, unconstrained, the optimizer created a wing with a maximum deflection of almost 3.5 meters (with a half-span of 5.5 m), which would significantly alter the lift and drag distributions. This model constrains it to the proper amount of deflection.

The chord length distribution reaches the lower bound at the tips; however, it is not a rectangular wing, appearing more elliptical in shape. This implies that the minimum lift force, maximum deformation, and minimum lift-to-induced-drag ratio constraints were required to prevent the optimizer from creating a wing with a constant chord length equal to its lower bound (0.2 m). It would do this for both the chord and the height distributions without the constraints, as it would minimize the mass. These constraints force it to meet real-world expectations.

The twist distribution shows the tips of the wing as being twisted downwards. Due to `VortexLattice's` inability to calculate stall, this optimization does not consider stall characteristics. This distribution is simply a result of the constraints placed upon it, specifically the lift-to-induced-drag ratio, the minimum lift force, and the maximum bending moment stress. The monotonicity constraint successfully forces the twist to decrease, yielding a distribution from $-5°$ to $+5°$, minimizing the root bending moments by unloading the tips, where the moment arm is longest. This reduces the material needed to meet the 120 MPa

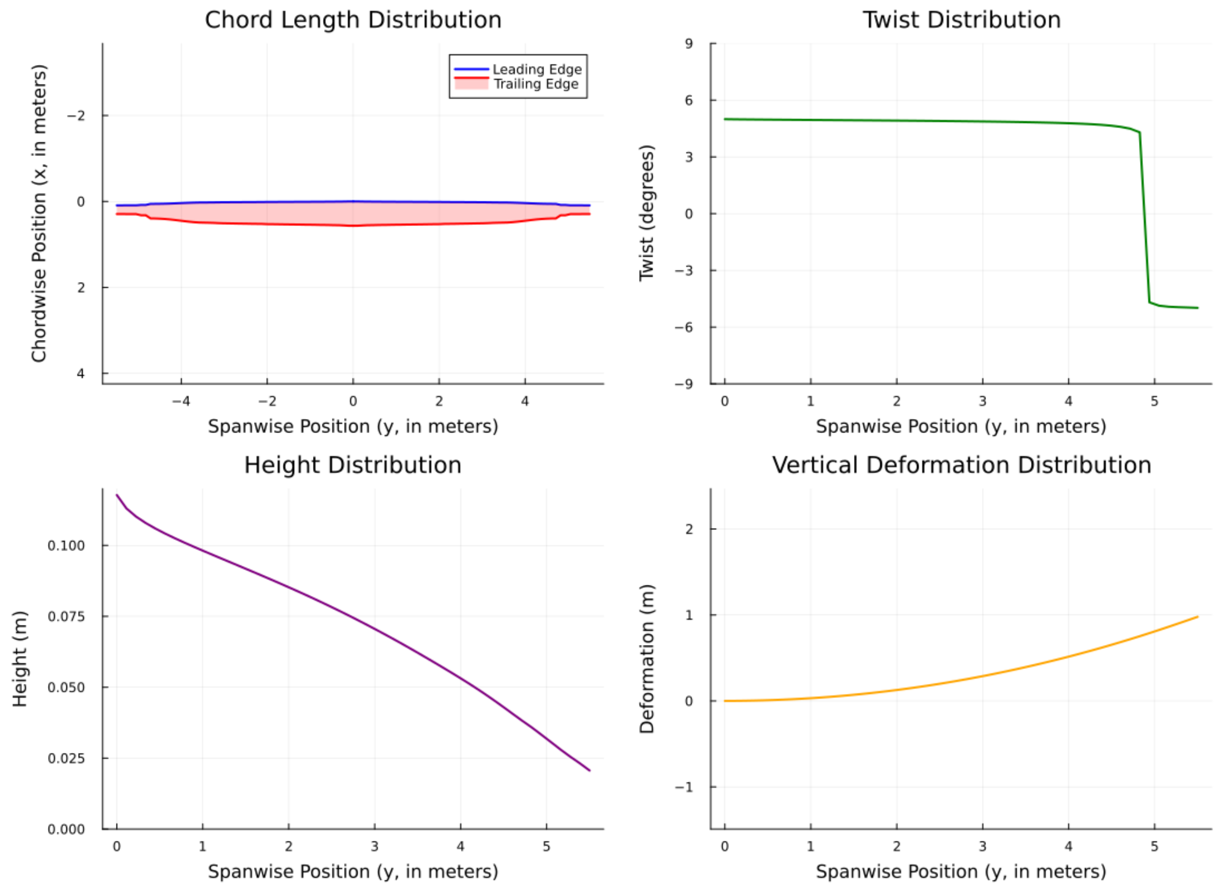

Figure 4: Results of the trade study, with the specific values listed below and in Appendix A.

stress limit. It is likely caused by a combination of these constraints, as **the maximum bending stress is 58.56 MPa**, resulting in a safety factor of 4.7; **the lift-to-induced drag ratio is 73.91**, and it barely surpasses the minimum lift force constraint. While this twist distribution is "optimal", a real twist distribution might have a more linear decrease in slope throughout the span to ensure that stall always occurs towards the roots instead of the tips. Note that this rapid and "idealistic" decrease in twist is also highly difficult to manufacture.

Shear forces are not a concern as the wing cannot experience a shear force greater than that of the combined lift and drag forces, and Al 6061-T6 has a shear strength of 207 MPa[5]. It will never reach that limit as the minimum required lift is only 11,281 N. Most wing designs have a lift-to-total-drag ratio of approximately 10:1. Even with a lift force of 20,000 N, that would imply that the maximum shear force the wing would experience would be 22,000 N, making it highly unlikely that shear could ever be the source of structural failure.

[5]https://www.matweb.com/search/datasheet.aspx?MatGUID=b8d536e0b9b54bd7b69e4124d8f1d20a

Comparing these results with the geometry of the Cessna Skyhawk verifies that this optimization provides valid results. The Cessna Skyhawk has a similar chord length distribution, and the height distribution is also similar, ranging from 11-16 cm. The optimization's results provide a similar root height, but a much smaller tip height[6]. The twist was unable to be compared to the Skyhawk's due to an inability to access its twist distribution. The mass of the Skyhawk's wing, however, is approximately 100 kg, implying that wing struts, internal structural beams, and a decreased wall thickness may allow for a smaller mass (as discussed in *5.1 (Deformation)*). In short, although this optimization yielded valid results, it may yet be improved upon to create a more accurate solution

[6]https://www.iieta.org/download/file/fid/76698: :text=Table%201.,%2D2.07o